

```

1. read the sample csv file
import numpy as np
import pandas as pd
data=pd.read_csv('/content/sample_data/california_housing_test.csv')
print(data)

```

	longitude	latitude	housing_median_age	total_rooms
total_bedrooms \				
0	-122.05	37.37	27.0	3885.0
661.0				
1	-118.30	34.26	43.0	1510.0
310.0				
2	-117.81	33.78	27.0	3589.0
507.0				
3	-118.36	33.82	28.0	67.0
15.0				
4	-119.67	36.33	19.0	1241.0
244.0				
...	...	...	...	...
...				
2995	-119.86	34.42	23.0	1450.0
642.0				
2996	-118.14	34.06	27.0	5257.0
1082.0				
2997	-119.70	36.30	10.0	956.0
201.0				
2998	-117.12	34.10	40.0	96.0
14.0				
2999	-119.63	34.42	42.0	1765.0
263.0				

	population	households	median_income	median_house_value
0	1537.0	606.0	6.6085	344700.0
1	809.0	277.0	3.5990	176500.0
2	1484.0	495.0	5.7934	270500.0
3	49.0	11.0	6.1359	330000.0
4	850.0	237.0	2.9375	81700.0
...	...	...	...	...
2995	1258.0	607.0	1.1790	225000.0
2996	3496.0	1036.0	3.3906	237200.0
2997	693.0	220.0	2.2895	62000.0
2998	46.0	14.0	3.2708	162500.0
2999	753.0	260.0	8.5608	500001.0

[3000 rows x 9 columns]

2.read the last 5 values of the csv file

```
data.tail()
```

	longitude	latitude	housing_median_age	total_rooms
total_bedrooms \				
2995	-119.86	34.42	23.0	1450.0
642.0				
2996	-118.14	34.06	27.0	5257.0
1082.0				
2997	-119.70	36.30	10.0	956.0
201.0				
2998	-117.12	34.10	40.0	96.0
14.0				
2999	-119.63	34.42	42.0	1765.0
263.0				

	population	households	median_income	median_house_value
2995	1258.0	607.0	1.1790	225000.0
2996	3496.0	1036.0	3.3906	237200.0
2997	693.0	220.0	2.2895	62000.0
2998	46.0	14.0	3.2708	162500.0
2999	753.0	260.0	8.5608	500001.0

1. read first 5 values of the csv file
`data.head()`

	longitude	latitude	housing_median_age	total_rooms
total_bedrooms \				
0	-122.05	37.37	27.0	3885.0
661.0				
1	-118.30	34.26	43.0	1510.0
310.0				
2	-117.81	33.78	27.0	3589.0
507.0				
3	-118.36	33.82	28.0	67.0
15.0				
4	-119.67	36.33	19.0	1241.0
244.0				

	population	households	median_income	median_house_value
0	1537.0	606.0	6.6085	344700.0
1	809.0	277.0	3.5990	176500.0
2	1484.0	495.0	5.7934	270500.0
3	49.0	11.0	6.1359	330000.0
4	850.0	237.0	2.9375	81700.0

1. read last 2 data from csv file
`data.tail(2)`

	longitude	latitude	housing_median_age	total_rooms
total_bedrooms \				
2998	-117.12	34.10	40.0	96.0
14.0				

```
2999      -119.63      34.42      42.0      1765.0
263.0
```

```
      population  households  median_income  median_house_value
2998          46.0          14.0          3.2708          162500.0
2999          753.0          260.0          8.5608          500001.0
```

1. read only last 2 data from csv file
`data.head(2)`

```
      longitude  latitude  housing_median_age  total_rooms
total_bedrooms \
0      -122.05      37.37      27.0      3885.0
661.0
1      -118.30      34.26      43.0      1510.0
310.0
```

```
      population  households  median_income  median_house_value
0          1537.0          606.0          6.6085          344700.0
1           809.0          277.0          3.5990          176500.0
```

1. get the details of the csv sample csv file
`data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   longitude              3000 non-null  float64
1   latitude               3000 non-null  float64
2   housing_median_age     3000 non-null  float64
3   total_rooms            3000 non-null  float64
4   total_bedrooms         3000 non-null  float64
5   population             3000 non-null  float64
6   households              3000 non-null  float64
7   median_income          3000 non-null  float64
8   median_house_value     3000 non-null  float64
dtypes: float64(9)
memory usage: 211.1 KB
```

1. print the total number of rows and column in the csv file
`data.shape`

```
(3000, 9)
```

```
data=data.append(data)
data.shape
```

```
(6000, 9)
```

1. remove all the duplicates value from the csv file and print the total number of rows and column

```
data=data.drop_duplicates()
data.shape
```

```
(3000, 9)
```

1. change the name of the column of the sample file

```
col_names = ['long',
             'lat',
             'HMA',
             'TROOMS',
             'TBEDROOMS',
             'P',
             'HH',
             'MI',
             'MHV' ]
```

```
data=pd.read_csv('/content/sample_data/california_housing_test.csv',
names=col_names)
print(data.head())
```

	long	lat	HMA	TROOMS
TBEDROOMS \				
0	longitude	latitude	housing_median_age	total_rooms
total_bedrooms				
1	-122.050000	37.370000	27.000000	3885.000000
661.000000				
2	-118.300000	34.260000	43.000000	1510.000000
310.000000				
3	-117.810000	33.780000	27.000000	3589.000000
507.000000				
4	-118.360000	33.820000	28.000000	67.000000
15.000000				

	P	HH	MI	MHV
0	population	households	median_income	median_house_value
1	1537.000000	606.000000	6.608500	344700.000000
2	809.000000	277.000000	3.599000	176500.000000
3	1484.000000	495.000000	5.793400	270500.000000
4	49.000000	11.000000	6.135900	330000.000000

1. create a new csv file and write any 3 datqa into the csv file

```
import pandas as pd
cities = pd.DataFrame([[ 'Sacramento', 'California'], [ 'Miami',
'Florida']], columns=[ 'City', 'State'])
cities.to_csv('cities.csv')
df = pd.read_csv('cities.csv')
print(df)
```

	Unnamed: 0	City	State
0	0	Sacramento	California
1	1	Miami	Florida

```
import pandas as pd
cities = pd.DataFrame([[ 'Sacramento', 'California'], [ 'Miami',
'Florida']], columns=[ 'City', 'State'])
cities.to_csv('cities.csv', index=False)
df = pd.read_csv('cities.csv')
print(df)
```

	City	State
0	Sacramento	California
1	Miami	Florida

1. change the column name of the csv file created

```
import pandas as pd
cities = pd.DataFrame([[ 'Sacramento', 'California'], [ 'Miami',
'Florida']], columns=[ 'City', 'State'])
new_column_names = [ 'City_Name', 'State_Name']
cities.to_csv('cities.csv', index=False, header=new_column_names)
df = pd.read_csv('cities.csv')
print(df)
```

	City_Name	State_Name
0	Sacramento	California
1	Miami	Florida

1. enter one unknown value to the csv file created

```
import pandas as pd
cities = pd.DataFrame([[ 'Sacramento', 'California'], [ 'Miami',
'Florida'], [ 'Washington DC', pd.NA]], columns=[ 'City', 'State'])
cities.to_csv('cities.csv', index=False, na_rep='Unknown')
df = pd.read_csv('cities.csv')
print(df)
```

	City	State
0	Sacramento	California
1	Miami	Florida
2	Washington DC	Unknown

1. create a json file and write into json file

```
import json
dictionary = {
    "name": "sathiyajith",
    "rollno": 56,
    "cgpa": 8.6,
    "phonenumber": "9976770500"
}
json_object = json.dumps(dictionary, indent=4)
```

```
with open("sample.json", "w") as outfile:  
    outfile.write(json_object)
```

1. read the created json file

```
import json  
with open('sample.json', 'r') as openfile:  
    json_object = json.load(openfile)
```

```
print(json_object)  
print(type(json_object))
```

```
{'name': 'sathiyajith', 'rollno': 56, 'cgpa': 8.6, 'phonenum':  
'9976770500'}  
<class 'dict'>
```